Jan Tauberschmidt

Curriculum Vitae

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GitHub ▪ Google Scholar ▪ arXiv ▪ OpenReview ▪ LinkedIn



Research Profile

My research focuses on developing machine learning methods and tools for engineering and sciences, with the long-term goal of automating costly iterative development processes. So far, I have explored efficient surrogate modeling and physics-constrained generative models for proposing plausible candidate solutions. Ultimately, I want to combine these approaches within closed-loop, multi-fidelity black-box optimization to discover diverse, high-performing designs while minimizing needed real-world experiments.

Practical Experience

since 11/2024 **Researcher at DFKI**, *Department Data Science and its Applications*
Working on scienfic machine learning and generative models.

04/2024 – 10/2024 **Master Thesis Student at Porsche Motorsport**
Developed a reinforcement learning framework for imitating professional human race car drivers.

09/2023 – 03/2024 **Research Assistant at DFKI**, *Department Data Science and its Applications*
Worked on spatio-temporal anomaly detection with semantic latent-space explanations.

09/2022 – 10/2023 **Research Assistant at Fraunhofer ITWM**, *Department for System Analysis, Prediction and Control*
Helped to develop a model-predictive controller for demand-side energy managment of factories.

03/2022 – 09/2022 **Internship at Porsche Motorsport**
Toolchain development for vehicle science and race engineers in top-level motorsport (LDMh and Formula-E).

09/2020 – 09/2021 **Team Leader, Formula Student**, *Kaiserslautern Racing Team (KaRaT)*
Led the overall development of an all-wheel-drive electric race car, technical lead for mechanics.

10/2018 – 10/2020 **Felix-Klein Modeling Weeks**, *Fraunhofer ITWM*
Focused group projects applying mathematical modeling to real-world problems. Topics: 10/2018: Quantification of uncertainties in district heating networks; 10/2019: Simulation and parameter analysis in thermal ablation of tumors; 10/2020: Analysis of stitching patterns in the production of nonwovens

Education

since 11/2025 **PhD Student**, *RPTU Kaiserslautern*
Supervised by Sebastian J. Vollmer, (informally) co-supervised by Andrew B. Duncan (Imperial College London)

10/2022 – 12/2024 **MSc Mathematics**, *RPTU Kaiserslautern*
Specialization Systems and Control, average grade: 1.0
Thesis: A Domain-informed Reinforcement Learning Race Driver Model

10/2018 – 03/2023 **BSc Mathematics**, *TU Kaiserslautern*
Specialization Industrial Mathematics, average grade: 1.6
Thesis: Model-based Torque Control of an Electric All-wheel Driven Race Car

09/2010 – 06/2018 **High School (Abitur)**, *Erasmus-Widmann-Gymnasium*
Schwäbisch Hall, average grade: 1.0

Awards

10/2018 – 09/2020 **Felix-Klein-Scholarship for Mathematicians**
Fraunhofer ITWM and Department of Mathematics Kaiserslautern.

06/2018 **A-Level Award for Mathematics**
For outstanding performance in the final exam in mathematics.

06/2018 **Other A-Level Awards**
Awards for year's best performance in Physics, German, Latin, Social Studies, Music; further distinctions for scholastic commitment and social engagement.

Publications

2026 **Physics-Constrained Fine-Tuning of Flow-Matching Models for Generation and Inverse Problems**
Jan Tauberschmidt, Sophie Fellenz, Sebastian J. Vollmer, and Andrew B. Duncan.
International Conference on Learning Representations (ICLR), 2026.

2026 **Heavy-tailed Physics-Informed Neural Networks**
Jephte Abijuru, Mayank Nagda, **Jan Tauberschmidt**, Phil Ostheimer, Sebastian J. Vollmer, Stephan Mandt, Marius Kloft, and Sophie Fellenz.
International Conference on Machine Learning (ICML), 2026.

2026 **Learning PDE Dynamics between Submanifolds Using Green's Observation Operators**
Jan Tauberschmidt, Jephte Abijuru, Samuel Okon, Naukshatro Bose, Sophie Fellenz, Marius Kloft, Jonas Latz, and Sebastian J. Vollmer.
arXiv preprint arXiv:2610.01697, 2026.

2026 **Walking the Score Manifold: Continuous-time Generative Dynamics on Learned Data Manifolds**
Jan Tauberschmidt, Brian B. Moser, Stanislav Frolov, Andreas Dengel, Andrew B. Duncan, and Sebastian J. Vollmer.
arXiv preprint arXiv:2609.17901, 2026.

2026 **Behavior-Constrained Reinforcement Learning with Receding-Horizon Credit Assignment for High-Performance Control**
Siwei Ju, **Jan Tauberschmidt**, Oleg Arenz, Peter van Vliet, and Jan Peters.
arXiv preprint arXiv:2604.03023, 2026.